Vector Search with FAISS

What FAISS is

FAISS is a library for efficient similarity search and clustering of dense vectors, described by Johnson, Douze and Jegou. It runs on CPU and optionally on GPU, and it offers many index types that trade accuracy, speed and memory.

Exact indexes

IndexFlatL2 and IndexFlatIP perform exact brute-force search using Euclidean distance and inner product respectively. The cost of a query grows linearly with the number of vectors times the vector dimension. For corpora with up to a few hundred thousand vectors, exact search is usually fast enough and has the advantage of returning the true nearest neighbours with no tuning.

Approximate indexes

IndexIVFFlat partitions the vectors into nlist clusters and, at query time, only scans the nprobe closest clusters, which is faster but may miss some true neighbours. IndexHNSWFlat builds a layered graph of neighbours and is a common choice when low latency matters on large collections. Approximate indexes require choosing parameters and measuring recall against an exact index.

Persistence

A FAISS index can be saved to disk with write_index and loaded again with read_index. FAISS stores only vectors and their positions, not the text or metadata. A RAG system must therefore keep a separate mapping from vector position to chunk text, source file and page number, and must keep that mapping in sync with the index file.

Metadata filtering

FAISS flat indexes do not filter by metadata natively. A simple approach is to retrieve more candidates than needed and discard those that do not match the filter, for example retrieving 5 times top-k candidates and then keeping only chunks from the requested document.

Keeping the index up to date

When a source document changes, the stored vectors for that document become stale. A reproducible strategy is to record a hash of every source file in a manifest and rebuild the index whenever the set of hashes changes. Duplicate chunks with identical text should be removed before indexing so that they do not crowd the top-k results.